Inorganic Chemistry CUME

Nov-16-2018

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

(1). Based on the Latimar Diagram of Mn (see below), calculate all these steps' standard electrochemical potentials, and predict which step(s) can be reversible. (20')

E^o_A (V)	0.564		2.27		0.95		1.51		-1.18	
MnO_4^-	→	MnO_4^{2-}	→	MnO_2	→	Mn^{3+}	→	Mn^{2+}	→	Mn

(2). Among these Mn oxidation states, which one(s) can show the Jahn-Teller Effect? Please use a simple sketch to explain each such effect. (20')

(3). A zeolite-like microporous crystal structure of the MnO_2, called hollandite that's a good battery material, has a [2 x 2] tunnel structure that's formed by the building blocks of (MnO_6) via corner- and edge-sharing. (i). Draw the structure of the tunnel. (ii) Estimate the largest size of a hydrated cation (or molecule) that can be adsorbed into the tunnel. (20')

(4). Recently, several chemistry seminars were on the topic of *metal organic framework* (or MOF) that is a new forefront in the modern inorganic chemistry. (i) Use the (MnO_6) cluster and a linear organic ligand (or molecule) of your choice to design and construct a new 3D-framework structure, and (ii) draw the 3D-structure and predict the crystal lattice type. (20')

(5). By replacing the (MnO_6) with the (MnO_4) cluster and using a linear ligand (or molecule) of your choice, (i) propose an experimentally doable route to the synthesis of the 3D-framework, and (ii) use some commonly seen methods to characterize the crystal lattice type, pore size, and surface area. (20')

(6). If connecting the (MnO_4) clusters purely (i.e. not using any other ligands or molecules) via corner-sharing, draw the product's 3D-structure. If thus-connecting the (MnO_6) clusters, draw the product's 3D-structure. (20')

(7). For checking the oxidation state of Mn^{n+} in the (MnO_4) and/or (MnO_6) clusters, how many and what methods you can use and what data can help you confirm the oxidation state? (20')

(8). For checking what small molecules would be adsorbed in the porous structures, how many and what methods can you use to get what data? (20')

(9). If you plan to make a new type of the electrically conducting MOF-like framework, what ligand or molecule or moiety can you use, and what data from what methods can help you confirm the electrical conductivity? (20')

(10). If small drug-molecules be adsorbed into the framework, what data from what methods can help you understand (i) what percentage of the total pore-surface be covered by the drug molecules and (ii) whether the drug molecules be only on the pores' outer surface, or inner surface, or all surfaces of a micrometer-sized solid of the very framework. (20')